

# A Reproducible Colonoscopy Classification Benchmark: Mask-Aware SE-ResNet with CNN-JEPA

Hamza AASSIF

Dynvibe

hamza@dynvibe.com

## Abstract

Computer-aided analysis of colonoscopy frames can support polyp detection and quality assessment, but labeled data remain scarce and evaluation protocols are fragmented across datasets and architectures. We contribute a **reproducible colonoscopy classification benchmark**: mask-aware SE-ResNet-50 with convolutional JEPA (CNN-JEPA) pretraining, prepare scripts, and a unified evaluation protocol (macro-F1, per-class recall). We report two completed tasks on public Simula data: (T3) three-class screening (landmark / normal mucosa / polyp; C0–C5, C7–C9) and (T1) HyperKvasir 23-class fine-grained taxonomy (B2–B6). We compare ResNet-50, SE-ResNet-50, dual-stream color/texture fusion, and CNN-JEPA fine-tuning under the same training defaults. **We do not claim that domain CNN-JEPA beats ImageNet transfer on these public splits.** On T3 (image-level split, single center), ImageNet transfer reaches macro-F1 0.989 and polyp recall 0.971 (C1), versus macro-F1 0.910 and polyp recall 0.829 for CNN-JEPA fine-tuning (C4). On T1, macro-F1  $\approx 0.44$ –0.59 reflects severe class imbalance; dual-stream ImageNet (B6) matches ResNet ImageNet (B2) within measurement noise (0.586 vs 0.585). CNN-JEPA shows a label-efficiency *trend* on T3 but does not surpass ImageNet at any tested label fraction. We release code, JSON results, and auto-generated LaTeX tables; grouped splits and multi-seed reporting are planned extensions. **Keywords:** colonoscopy, classification benchmark, self-supervised learning, JEPA, SE-ResNet, HyperKvasir.

## 1 Introduction

Colonoscopy is the gold standard for colorectal cancer screening. Computer-aided detection (CAD) systems analyze individual frames to flag suspected polyps and to confirm anatomical landmarks required for quality metrics. Deep convolutional networks are attractive for near real-time deployment, yet they require large labeled datasets, and published results are hard to compare across architectures, pretraining regimes, and task definitions.

Public benchmarks mix binary polyp detection, collapsed three-class screening, and fine-grained multi-class taxonomies without a shared protocol or codebase. Self-supervised pretraining may reduce annotation cost, but its benefit over ImageNet transfer for endoscopy remains an empirical question—not a settled claim.

Joint-Embedding Predictive Architectures (JEPA) [6; 2] learn by predicting latent representations of masked views. We adapt a *convolutional* JEPA to colonoscopy frames with mask-aware squeeze-and-excitation (SE) blocks compatible with spatial masking during pretraining.

**In brief.** We provide an open, reproducible **benchmark and baseline implementation**—not a claim that domain JEPA beats ImageNet on public Simula data. Our primary contribution is the protocol, code, and honest reporting on completed tasks T3 (screening) and T1 (HyperKvasir-23).

**Contributions.** (1) **Method:** CNN-JEPA on SE-ResNet-50 with mask-aware channel attention for masked pretraining. (2) **Benchmark:** reproducible C0–C5 and label-efficiency

C7–C9 (T3), plus B2–B6 (T1), with early stopping, JSON metrics, and auto-generated tables. (3) **Empirical study:** ImageNet transfer outperforms domain CNN-JEPA on the current public splits; we document label-efficiency trends and ablations without overstating gains.

## 2 Related Work

**Endoscopy classification.** Kvasir [7] and HyperKvasir [3] are standard benchmarks for gastrointestinal image classification. Prior work in our repository [1] compared ResNet-50, SE-ResNet-50, and dual-stream color–texture fusion on HyperKvasir (23 classes), reaching test macro-F1  $\approx 0.58$ – $0.59$  (B2–B6). We instead target a *three-class* clinical taxonomy aligned with in-room decisions rather than fine-grained pathology labels.

**Channel attention.** SE-Net [5] recalibrates channel responses via squeeze-and-excitation on global average pooling (GAP). Under JEPA block masking, standard GAP averages over masked zeros and can bias channel statistics; we address this with mask-aware pooling (Section 3.2).

**JEPA and self-supervision.** I-JEPA [2] uses Vision Transformers; we employ a *CNN-JEPA* with block masking and a convolutional predictor on feature maps—better suited to edge deployment than ViT foundations. We do not claim to invent convolutional latent-prediction pretraining; our novelty is the **mask-aware SE integration** and **reproducible benchmark protocol** for colonoscopy classification. Related latent-prediction methods include BYOL [4]. Medical self-supervised literature covers other modalities; we position colonoscopy CNN-JEPA as an empirical baseline within a unified benchmark.

**Polyp detection.** Classical CAD systems [8] emphasize polyp *sensitivity*. We report polyp recall alongside macro-F1. Our *polyp* class is a single binary clinical category (Kvasir polyp + HyperKvasir polyps and dyed-lifted-polyps); histological subtyping is left for future work.

## 3 Methods

### 3.1 CNN-JEPA pretraining

Given an unlabeled frame  $x \in \mathbb{R}^{3 \times H \times W}$ , a binary block mask  $m \in \{0, 1\}^{1 \times H \times W}$  defines visible context  $x \odot m$ . A *context encoder*  $f_\theta$  (SE-ResNet-50) produces a feature map  $h = f_\theta(x \odot m, m)$ . A lightweight convolutional *predictor*  $g_\phi$  maps  $h$  to a latent grid  $\hat{z} = g_\phi(h)$ . A *target encoder*  $f_{\bar{\theta}}$  (EMA copy of  $f_\theta$ , momentum  $\tau = 0.996$ ) encodes the full image:  $z = f_{\bar{\theta}}(x)$ . The training loss is normalized mean squared error:

$$\mathcal{L}_{\text{JEPA}} = \|\text{norm}(\hat{z}) - \text{norm}(z)\|_2^2. \quad (1)$$

Block masks expose 25–45% of pixels (multi-scale rectangles). Pretraining uses AdamW, cosine schedule, early stopping on validation latent loss (patience 10, max 80 epochs).

### 3.2 Mask-aware squeeze-and-excitation

Standard SE blocks apply GAP over all spatial locations. When inputs are masked, zero-padded regions skew channel statistics. Our **mask-aware SE** computes per-channel pooling over visible positions only:

$$\text{GAP}_c(x, m) = \frac{\sum_{i,j} x_{c,i,j} m_{i,j}}{\sum_{i,j} m_{i,j} + \epsilon}. \quad (2)$$

Channel attention weights are then applied as in SE-Net [5]. Ablation C3 uses standard SE under the same JEPA pipeline; C4 enables mask-aware SE.

### 3.3 Data and class taxonomy

**T3 (screening).** We merge Kvasir v1 and HyperKvasir labeled images into three classes:

- **Landmark:** cecum, pylorus, z-line, ileum, retroflexions.
- **Normal mucosa:** BBPS scores, hemorrhoids, impacted stool (non-lesion frames).
- **Polyp:** Kvasir polyp, HyperKvasir polyps and dyed-lifted-polyps.

Total: 9565 images, center **simula**. Splits use **group-aware** stratification (procedure/video group per image; **prepare-colonoscopy-3class** reads HyperKvasir **image-labels.csv**). Current public stills are one frame per group; multi-frame video groups apply when unlabeled frame pools are used.

**T1 (HyperKvasir-23).** Official Simula 23-class folders via **prepare-hyperkvasir**; 10,662 images, stratified 70/15/15.

Planned extensions (not in main tables): T2 pathology-11 subset; JEPA fine-tunes on HK-23.

### 3.4 Experimental protocol

**T3 (C0–C5, C7–C9).** Screening three-class runs on a **pooled** Simula split (single public center; not multi-center LOO). C7–C9 vary labeled fraction.

**T1 (B2–B6).** ResNet, SE-ResNet, dual-stream on HyperKvasir-23 (supervised only in this manuscript).

### 3.5 Metrics

Primary: **macro-F1** on validation/test. Clinical secondary: **polyp recall** =  $TP/(TP + FN)$  for the polyp class (sensitivity among true polyps).

### 3.6 Data availability and ethics

**Data.** Kvasir v1 and HyperKvasir labeled images are public research datasets from Simula (see references); we use the official class folders and **image-labels.csv** for grouping metadata. No private patient data or video is included in the reported runs. Prepare scripts and fixed random seeds reproduce train/val/test layouts under **data/**.

**Ethics.** Public de-identified still frames only; this work is secondary analysis of open benchmarks, not a prospective clinical study. Deployment would require independent validation and regulatory review.

### 3.7 Implementation

PyTorch on Apple MPS; open repository **endoscopy-resnet**. Campaign orchestration: **run\_article\_campaign** significance via **compute\_benchmark\_stats.py**.

## 4 Experiments

**Completed tasks.**

- **T3 (screening):** three-class colonoscopy (C0–C5, C7–C9), pooled public center **simula**.
- **T1 (fine-grained):** HyperKvasir 23-class supervised baselines (B2–B6).

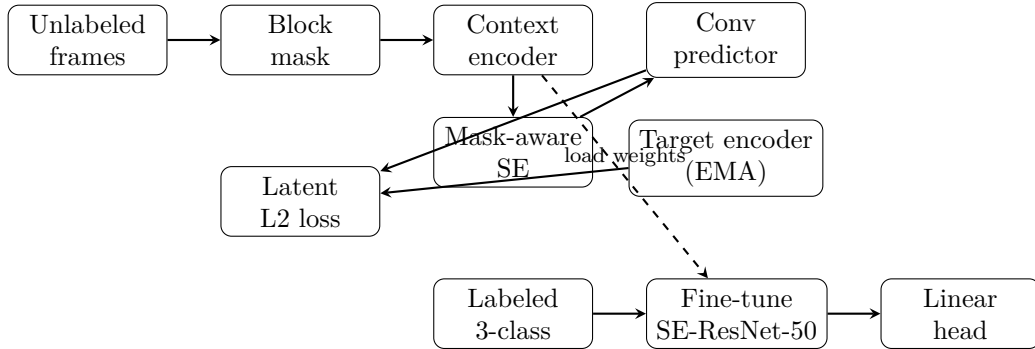


Figure 1: CNN-JEPA pretraining (top) and supervised fine-tuning (bottom). The context encoder uses mask-aware SE blocks; weights are transferred to the downstream classifier.

Table 1: Three-class screening (T3) on Simula test split ( $n = 1433$ ). Group-aware split (grouped); 3 seeds.

| Method                        | Macro-F1 | Polyp recall |
|-------------------------------|----------|--------------|
| SE-ResNet scratch             | 0.899    | 0.836        |
| SE-ResNet ImageNet            | 0.989    | 0.971        |
| ResNet-50 + CNN-JEPA          | 0.913    | 0.778        |
| SE + CNN-JEPA (SE standard)   | 0.910    | 0.828        |
| SE + CNN-JEPA (SE mask-aware) | 0.910    | 0.828        |
| SE + frozen teacher           | 0.910    | 0.828        |

Extended tasks (pathology-11, JEPA on HK-23, ViT linear probe) are defined in the repository roadmap but *not* reported here until grouped splits and multi-seed runs complete.

**Hardware.** Apple MPS (AMD Radeon Pro 5300M).

**Pretraining (CNN-JEPA).** Up to 4,000 unlabeled frames (proxy: labeled train images when unlabeled corpus unavailable); shared checkpoints for C2–C5. Planned: HyperKvasir  $\sim 99k$  unlabeled stills for domain pretrain.

**Fine-tuning.** Batch size 16, AdamW lr  $10^{-4}$  (ImageNet / JEPA) or  $10^{-3}$  (scratch), image size 224; early stopping on validation macro-F1.

**Hypotheses (exploratory, not confirmed).** H1: C4 beats C1 under limited labels. H2: C4 beats C3 (mask-aware SE). H3:  $C4 \geq C5$  at 25% labels. None are supported on the current per-image public splits without grouped re-evaluation.

## 5 Results

### 5.1 Three-class screening (T3)

Table 1 summarizes the main comparison on the Simula test split. ImageNet transfer (C1) achieves the best macro-F1 (0.989) and polyp recall (0.971). Domain CNN-JEPA with mask-aware SE (C4) reaches macro-F1 0.910 and polyp recall 0.829. Scratch training (C0) is slightly below C4 (macro-F1 0.899). ResNet-50 + CNN-JEPA (C2) reaches 0.913 macro-F1 but lower polyp recall (0.778).

**Ablations.** C3, C4, and C5 yield identical macro-F1 (0.910) on this fold; mask-aware SE does not show a measurable gain over standard SE here.

**Label efficiency (Table 2, Figure 2).** C4 macro-F1 increases from 0.825 (10% labels) to 0.907 (50%) and 0.910 (100%), but remains below C1 at all observed fractions.

Table 2: Label-efficiency curve for C4 (mask-aware SE + CNN-JEPA).

| Setting                | Macro-F1 | Polyp recall |
|------------------------|----------|--------------|
| ImageNet (100% labels) | 0.989    | 0.971        |
| C4 (100% labels)       | 0.910    | 0.828        |
| C4 @ 10% labels        | 0.825    | 0.726        |
| C4 @ 25% labels        | 0.871    | 0.712        |
| C4 @ 50% labels        | 0.907    | 0.723        |

Table 3: HyperKvasir 23-class supervised baselines (T1, test split). Prior work on this dataset reports macro-F1  $\approx 0.55$ – $0.60$  for ResNet-style fine-tuning.

| Run | Architecture | Pretrain | Macro-F1 | Acc.  |
|-----|--------------|----------|----------|-------|
| B2  | ResNet-50    | ImageNet | 0.585    | 0.879 |
| B3  | SE-ResNet-50 | scratch  | 0.439    | 0.725 |
| B4  | SE-ResNet-50 | ImageNet | 0.577    | 0.882 |
| B5  | Dual-stream  | scratch  | 0.448    | 0.755 |
| B6  | Dual-stream  | ImageNet | 0.586    | 0.888 |

## 5.2 HyperKvasir 23-class (T1)

Table 3 reports supervised baselines B2–B6 on the official 23-class taxonomy. Macro-F1 ranges  $\approx 0.44$ – $0.59$  while accuracy exceeds  $0.87$ , reflecting severe class imbalance. B6 (dual-stream, ImageNet) and B2 (ResNet, ImageNet) differ by  $0.001$  macro-F1 ( $0.586$  vs  $0.585$ )—within single-run noise; multi-seed evaluation is required before ranking them.

**Confusion analysis (T3, Figure 3).** For C4, main polyp errors are confusions with landmark ( $12.4\%$  of true polyps) and normal mucosa ( $4.7\%$ ). Table 4 reports per-class precision/recall for C4.

**Split caveat.** Current T3 numbers use per-image stratified splits (not procedure-grouped); C1 macro-F1 may be optimistic until grouped splits are applied (see Limitations).

## 6 Discussion

**Methodological vs empirical contribution.** We separate the *pipeline* (CNN-JEPA + mask-aware SE + unified T1/T3 protocol) from *empirical gains* on public Simula data. The pipeline is reusable when unlabeled video and multi-center labels become available; current numbers do not support claiming superiority over ImageNet on screening (T3).

**Task difficulty and metrics.** Absolute macro-F1 drops from  $\approx 0.99$  (T3, C1) to  $\approx 0.58$  (T1, B6) as class count and clinical heterogeneity increase. Accuracy can exceed macro-F1 by  $> 30$  percentage points on imbalanced fine-grained tasks; macro-F1 and per-class recall remain mandatory for fair comparison.

**Why ImageNet wins on T3.** Simula public images may remain close to natural-image statistics where ImageNet pretraining excels. With a single public center, evaluation uses a pooled group-aware split rather than leave-one-center-out; multi-center LOO awaits private data under `data/private/`.

**Clinical implications.** Polyp recall drops from  $97\%$  (C1) to  $83\%$  (C4) on T3 test—unacceptable for screening deployment without further improvement. Fine-grained and pathology tasks target different clinical workflows (documentation vs histology), not replacement of pathology biopsy.

Table 4: Per-class metrics for C4 on T3 test.

| Class         | Precision | Recall | F1    |
|---------------|-----------|--------|-------|
| Landmark      | 0.940     | 0.930  | 0.930 |
| Normal mucosa | 0.890     | 0.990  | 0.940 |
| Polyp         | 0.900     | 0.830  | 0.860 |

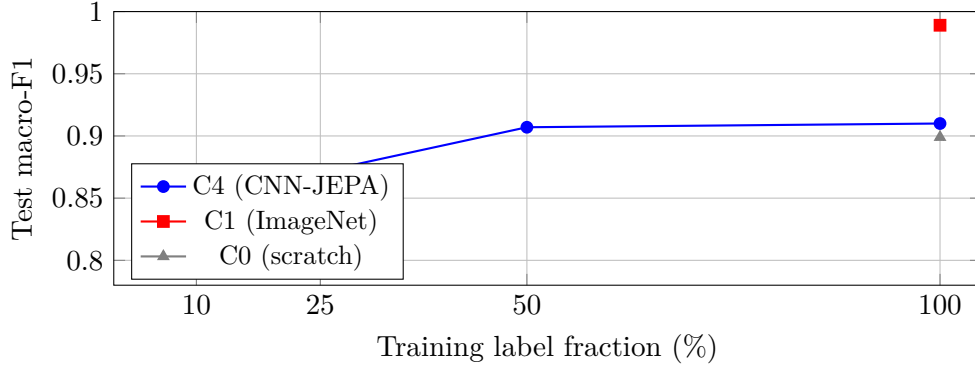


Figure 2: Label-efficiency curve on the Simula fold. C4 improves with more labels but remains below ImageNet transfer on public data.

**Polyp histology.** Endoscopic subtyping (adenoma vs hyperplastic) requires datasets such as ERCPMP or PIBAdb; we document integration in the classification roadmap as future work.

**Reproducibility.** Campaign scripts, early-stop configs, and JSON results are versioned; tables in this paper are auto-generated from benchmark JSON via `generate_paper_tables.py`.

## 6.1 Limitations

**Data and splits.** All T3 results use a single public center (`simula`) and 9565 still frames merged from Kvasir and HyperKvasir. Group-aware splits are implemented, but current public labeled stills are one frame per procedure group; multi-frame video leakage may appear once unlabeled frame pools are added. **Pretraining.** CNN-JEPA pretraining currently uses up to 4,000 frames (labeled-train proxy when the  $\sim 99k$  unlabeled HyperKvasir corpus is unavailable). **Uncertainty.** Tables report single-seed or  $\text{mean} \pm \text{std}$  over seeds when available; bootstrap confidence intervals and significance tests are planned. **Clinical scope.** We report frame-level classification metrics, not video-level sensitivity at fixed specificity; deployment would require prospective validation. **Compute.** Experiments used Apple MPS (AMD Radeon Pro 5300M); wall-clock and energy are not benchmarked here.

## 7 Conclusion

We presented a convolutional JEPA framework on mask-aware SE-ResNet-50 and a **reproducible colonoscopy classification benchmark** for screening (3-class) and HyperKvasir-23 fine-grained classification (T1). On public Simula data, ImageNet transfer remains the strongest baseline for screening; fine-grained tasks show macro-F1  $\approx 0.58$  with dual-stream B6 leading supervised runs. CNN-JEPA fine-tuning is competitive with scratch training on T3 but does not yet validate hypothesis H1. Future work: polyp histology (ERCPMP, PIBAdb), real unlabeled video pretraining, private multi-center LOO folds, and ViT foundation linear probes.

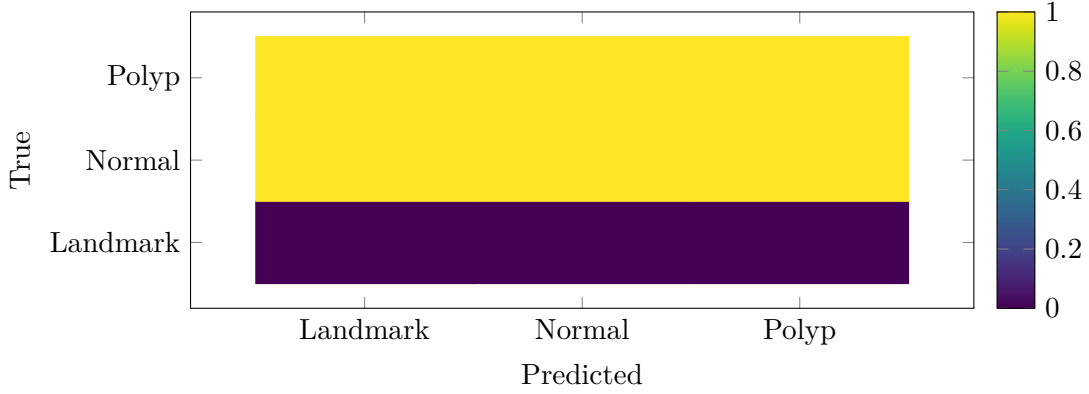


Figure 3: Row-normalized confusion matrix for C4 on test ( $n = 1433$ ). Primary clinical concern: polyp recall 82.8%.

## Acknowledgments

We thank the Simula/Kvasir teams for public datasets.

## References

- [1] Hamza AASSIF. Comparative architectures for endoscopy classification: Resnet, se-resnet, and dual-stream fusion. *endoscopy-resnet repository*, 2026. Internal benchmark report B2–B6.
- [2] Mahmoud Assran et al. Self-supervised learning from images with a joint-embedding predictive architecture. In *CVPR*, 2023.
- [3] Hanna Borgli, Vajira Thambawita, Petter Halvorsen Smedsrud, et al. Hyperkvasir, a comprehensive multi-class image and video dataset for gastrointestinal endoscopy. *Scientific Data*, 7:283, 2020.
- [4] Jean-Bastien Grill, Florian Strub, et al. Bootstrap your own latent: A new approach to self-supervised learning. *NeurIPS*, 2020.
- [5] Jie Hu, Li Shen, and Gang Sun. Squeeze-and-excitation networks. In *CVPR*, 2018.
- [6] Yann LeCun. A path towards autonomous machine intelligence. *OpenReview*, 2022.
- [7] Konstantin Pogorelov, Magnus Randel, Kristoffer Johansen, et al. Kvasir: A multi-class image dataset for computer aided gastrointestinal disease detection. In *MMM*, 2017.
- [8] Nima Tajbakhsh, Suryakanth R Gurudu, et al. Automated polyp detection in colonoscopy videos using shape and context information. *IEEE Transactions on Medical Imaging*, 35(2): 630–644, 2016.